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Design of e-commerce product price prediction model based on generative adversarial network with adaptive weight adjustment

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E-commerce platforms have amassed extensive transaction data, which serves as a valuable source for price prediction. However, the diversity of commodities poses challenges such as data imbalance, model overfitting, and underfitting. To address these issues, this paper presents an improved generative adversarial network model that integrates the strengths of Conditional Generative Adversarial Nets and the Wasserstein Generative Adversarial Network. By introducing Wasserstein scatter and removing Lipschitz constraints, we propose the CWGAN model to mitigate data imbalance and enhance the quality of generated samples. Furthermore, we incorporate Adaptive Weight Adjustment (AWA) and a differential evolution strategy, resulting in the Adaptive Weight Adjustment-Conditional Wasserstein Generative Adversarial Network (AWA-CWGAN) algorithm. This algorithm employs a neighborhood learning strategy to update the optimal individuals within subpopulations, thereby reinforcing the influence of elite individuals during population evolution. Additionally, dynamic weight adjustment based on sparsity is implemented to increase genetic diversity within the population. Experimental results demonstrate that the AWA-CWGAN algorithm achieves complete convergence with only 16–25% of the global evolutionary generations required by the standard differential evolutionary algorithm or the hybrid frog-leaping algorithm. Moreover, the AWA-CWGAN algorithm surpasses baseline methods in accuracy (88.8%), precision (88.81%), recall (89.255%), and F1 score (87.95%). These results indicate that the proposed approach significantly enhances the accuracy of e-commerce product price predictions, providing robust decision-making support for merchants.

Keywords Generative adversarial networks, Adaptive weight adjustment, Differential evolution, Price prediction

With the rapid advancement of e-commerce platforms, both consumers and merchants have experienced unparalleled convenience. However, these platforms face significant challenges, particularly in price forecasting and management. The frequent fluctuation of product prices requires merchants to devise effective pricing strategies to attract consumers and boost sales. A reliable price prediction model can enable merchants to respond swiftly to market dynamics and develop well-informed pricing strategies[1]. Additionally, e-commerce platforms have accumulated vast amounts of transaction data, including product attributes, sales volumes, and user behavior. This data serves as a rich foundation for developing robust price prediction models. Given the wide variety of products and the intense competition in e-commerce, accurately forecasting product prices has become a critical factor in enhancing platform competitiveness and meeting consumer demands.

In recent years, machine learning techniques have achieved remarkable success across various domains, particularly in price prediction. Generative Adversarial Networks (GANs)[2], a type of deep learning model comprising a generator and a discriminator, have emerged as powerful tools. The generator creates synthetic data samples, while the discriminator evaluates whether the samples are real or generated. GANs have achieved significant breakthroughs in fields such as image generation and data augmentation and have gradually extended their applications to areas like natural language processing and recommendation systems[3]. In the context of price prediction, GANs can generate synthetic price data resembling real-world prices, providing enhanced

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training datasets for price prediction models. By capturing the distribution of price data, GANs help models identify patterns and trends in price fluctuations, thereby improving prediction accuracy.

Adaptive Weight Adjustment (AWA) is a method for dynamically modifying model weights based on input data, model state, or task requirements to enhance model performance and flexibility[4]. In deep learning and machine learning, AWA is widely applied in domains such as multi-task learning, optimizer design, and attention mechanisms. In the context of price prediction, AWA enables the model to adjust its weights dynamically in response to factors like product attributes and market demand, allowing the model to better accommodate varying data distributions and task requirements. By capturing more useful information through this dynamic adjustment, AWA significantly improves the accuracy of price predictions.

Despite these advantages, e-commerce platforms often face severe data imbalance issues. For certain products, price data is plentiful, while for others, it is scarce. This imbalance can lead models to over-focus on the more abundant data during training, neglecting products with limited data, thereby reducing prediction accuracy and generalization capability. Moreover, when employing GANs and AWA techniques, avoiding overfitting and underfitting is critical. Overfitting enables the model to excel on training data but diminishes its performance on unseen data. Underfitting, conversely, results in the model failing to capture the underlying patterns in the data, thus compromising prediction accuracy[5].

To address these challenges, this paper improves GANs and integrates AWA with additional techniques to better capture the complex relationships between product prices and influencing factors. The specific contributions of this paper are as follows:

1. Improvement of GANs: Outlier detection for normal samples and the synthesis of new minority class samples are achieved using a Conditional Wasserstein Generative Adversarial Network (CWGAN). This approach ensures consistency in data distribution, addresses data imbalance, and increases sample diversity.
2. Neighborhood Learning Strategy in Local Search: A neighborhood learning strategy is employed during the local search process to update the optimal individuals within subpopulations. This strengthens the influence of elite individuals on group evolution, enhances the probability of discovering global optima, and accelerates algorithm convergence.
3. Global Search with AWA and Differential Evolution: Adaptive weight adjustment and differential evolution strategies are utilized during the global search process to genetically update mixed populations. The proposed AWA-CWGAN algorithm expands the dissemination of elite genes within populations, maintains genetic diversity, and enhances the algorithm's global optimization capabilities.

Related work

GANs represent a novel machine learning framework based on adversarial processes. Their primary purpose is to train generator models capable of learning the characteristics of real data distributions and generating realistic approximations. Literature[6] introduces the CGAN model, which extends the original GAN by incorporating conditional constraints through category labels. These labels guide the generator in producing more efficient data outputs, addressing some of the instability issues inherent in the original GAN model. Additionally, [6] proposes the DCGAN architecture, which replaces the traditional MLP structure with a Convolutional Neural Network (CNN) in the discriminator. This innovation significantly enhances model performance, as demonstrated through experimental validation.

Despite the promising potential of GANs and their variants in designing e-commerce product price prediction models—especially when combined with adaptive weight adjustment strategies to optimize prediction accuracy—several limitations persist. Traditional GANs often generate simplistic and redundant samples, failing to capture the complex distributional characteristics of e-commerce product price data. These characteristics include factors such as seasonal variations, promotional activities, and shifts in consumer behavior. Consequently, these shortcomings hinder the predictive accuracy and generalization performance of such models in real-world applications[7].

To address these challenges, advanced generative adversarial networks based on game theory have been proposed. These models introduce more intricate game mechanisms to generate diverse and realistic samples, thereby improving sample quality and addressing the prevalent data imbalance in e-commerce datasets. This approach not only enhances the realism of generated data but also significantly mitigates the imbalance issues that commonly plague e-commerce datasets, paving the way for more robust and accurate price prediction models[8].

The GAN model based on the attention mechanism, as proposed in literature[9], introduces attention mechanisms in the generator to extract key features and generate effective adversarial samples that retain crucial information. While this enhances the model's ability to process complex data, it increases implementation complexity and requires significant computational resources. Similarly, the multi-module integrated intrusion detection system introduced in literature[10] combines stacked self-encoders, Gaussian mixture models, and Wasserstein Generative Adversarial Network (WGAN) to address data imbalance. Although this system incorporates CNNs[11] and LSTM) networks[12] for improved classification accuracy and robustness, WGANs are prone to model collapse and training instability during sample generation. These issues arise from vanishing or exploding gradients during optimization and the difficulty of identifying a suitable distance metric to accurately quantify differences between generated and real data distributions, posing significant challenges and uncertainties in real-world deployment.

In designing e-commerce product price prediction models that leverage GANs and adaptive weight adjustment, it is essential to comprehensively evaluate the advantages and limitations of these methods and explore stable, efficient, and adaptive model architectures and training strategies. While GANs have demonstrated improvements in capturing data distributions, their performance in generating text within discrete

spaces remains limited, undermining their efficacy in price prediction for e-commerce products. Literature[13] proposes the SeqGAN model, which employs the policy gradient method from reinforcement learning[14] to address adversarial generation of discrete samples. Literature[15] introduces WGAN, which alleviates the gradient vanishing problem by calculating the Earth Mover's distance and adding noise to generated and real data.

Despite these advancements, significant challenges persist in applying GANs with adaptive weight adjustment to e-commerce price prediction. Literature[16] proposes a GAN structure utilizing LSTM as the generator and a multilayer perceptron (MLP) as the discriminator, demonstrating advantages in handling time-series data. However, the LSTM generator faces high computational complexity and gradient vanishing when processing long time-series data, reducing prediction accuracy and efficiency. Additionally, the MLP discriminator struggles to capture the complex nonlinear relationships in price data, limiting the model's learning capacity. The GAN-FD model proposed in literature[17], which employs LSTM as the generator and CNN as the discriminator, performs well in high-frequency stock market prediction by minimizing prediction error loss and directional error. However, its application to e-commerce price prediction is constrained because price fluctuations in e-commerce are influenced not only by time-series factors but also by complex elements such as market demand, promotional activities, and competitors' pricing strategies. CNNs may not be as effective as advanced graph neural networks or attention-based models in extracting global features relevant to these factors.

Literature[18] proposes a SeqGAN-based oversampling method to address data imbalance in stock review text mining, achieving improved classification accuracy. However, in e-commerce price prediction, data imbalance extends beyond text data to price data itself. For instance, there may be significantly more price data available for some products than others, potentially biasing the model towards commodities with abundant data. Moreover, while SeqGANs are effective for discrete-text generation tasks, they may not perform as well in continuous-value prediction tasks, which could impact the accuracy and stability of price predictions.

In conclusion, applying GANs to e-commerce price prediction requires careful consideration of data characteristics, model applicability, and task complexity to address their inherent limitations. Exploring innovative architectures and strategies is essential for overcoming these challenges and improving predictive performance.

Methodology

CWGAN

To address the issue of poor prediction performance caused by data imbalance in e-commerce product price prediction, this paper introduces the Conditional Wasserstein Generative Adversarial Network (CWGAN). By combining the strengths of Conditional Generative Adversarial Networks (CGAN)[19] and Wasserstein Generative Adversarial Networks (WGAN)[20], CWGAN mitigates model collapse and training instability during the learning process. This is achieved by improving the quality of newly generated samples for underrepresented classes.

CWGAN effectively captures the real data distribution in the dataset and generates synthetic samples for underrepresented categories based on predefined labels, thereby alleviating training set imbalances. The architecture of the CWGAN generator is illustrated in Fig. 1.

The Lipschitz constraint is removed at the input, and the Wasserstein scatter is introduced to obtain the objective function as:

$$L = E_{x-p_r}[D(x|y)] - E_{x-p_g}[D(x|y)] + kE_{x'-p_x}[||\nabla D(x')||^p] \tag{1}$$

where x' is obtained by randomly interpolating the samples on a straight line between the real sample x and the generated sample $G(z|y)$. p_r represents the distribution of samples generated by real data. p_g is the distribution of the attack samples generated by G. D represents the outputs of the discriminator for the real samples and the generated samples.

Let ε obey a uniform distribution on [0,1] to obtain Eq. (2):

$$x' = \varepsilon x + (1-\varepsilon)G(z|y) \tag{2}$$

G(z|y) represents the sample generated by the generator based on noise z under condition y. The objective function is obtained by combining the CWGAN objective number and WGAN as:

$$L_e = \min_G \max_D E_{x-p_r}[D(x|y)] - E_{z-p_z}[D(G(z|y)) + \lambda_1 L] \tag{3}$$

where λ_1 is the W scatter coefficient. The CWGAN model integrates the advantages of both CGAN and WGAN. By incorporating the Wasserstein divergence, CWGAN more effectively measures the distributional difference between generated and real data, while removing the Lipschitz constraint to simplify the training process. Equation (1) employs the Wasserstein divergence to quantify the distributional gap between real and generated samples via random interpolation along the line connecting them. Equations (2) and (3) combine the objective functions of CWGAN and WGAN to jointly optimize the performance of the generator and discriminator.

In the following, the data is equalized by intra-category expansion so that the generated samples are similar to the real samples in terms of feature distribution. The construction of the discriminator is shown in Fig. 2.

To meet the input format requirements of the CWGAN model, the original dataset is transformed into a 28 × 28 grayscale map by padding with zeros. Using an embedding layer and a fully connected layer, the sparse labels are converted into a 28 × 28 matrix containing feature data, which is then concatenated as input to the discriminator . Random noise following a Gaussian distribution is added to the original dataset. The random

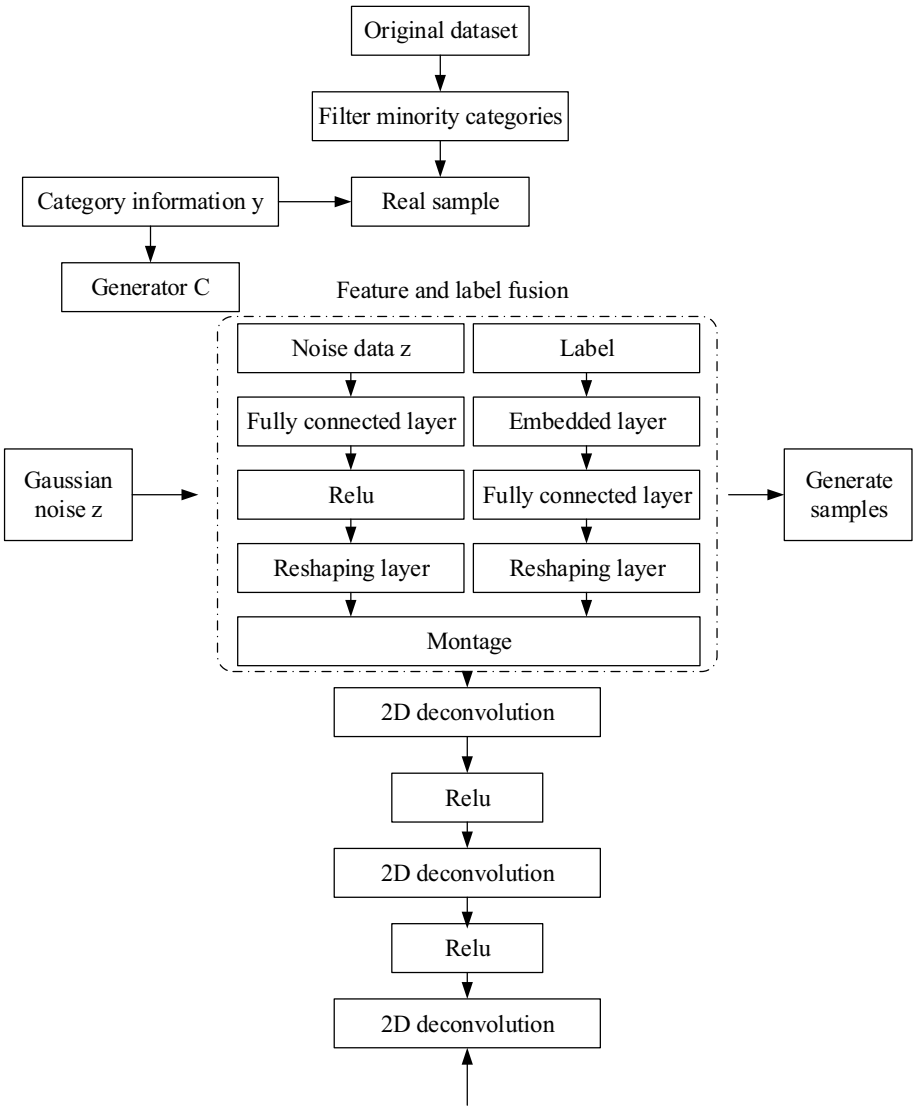

Fig. 1. Generator flow chart

noise vector , which also obeys a Gaussian distribution, undergoes a similar transformation into a 28×28 grayscale map. This transformed noise is concatenated with the transposed category labels and used as input to the generator .

The generator produces new attack samples, which are mixed with real samples and fed into the discriminator . The discriminator outputs the probability values that indicate whether the input samples belong to real samples () or generated pseudo-samples (). These probability values are finally converted into predicted category labels using an activation function, completing the prediction process.

AWA-CWGAN

For the predicted labels, the standard SFLA (Shuffled Frog-Leaping Algorithm)[21] is employed to perform a local search within each subpopulation. Subsequently, all subpopulations are remixed and exchanged through global information to identify the global optimal solution . During the subpopulation mixing process, beneficial information obtained from the local search is propagated to the entire population via individual fitness ranking. However, this approach does not generate new individuals or genetic variations, which limits the enrichment of the population's genetic diversity.

To address this limitation, a differential evolution (DE) strategy is introduced to perform genetic updates on the mixed population. The core principle of this algorithm involves generating temporary individuals based on the degree of individual differences within the population, followed by random recombination to facilitate population evolution. The mutation process is as follows:

$$V_{i,t} = X_{b,t} + F(X_{r1,t} + X_{r2,t} - X_{r3,t} - X_{r4,t}) \tag{4}$$

where $r_1 \neq r_2 \neq r_3 \neq r_r \neq i$. The crossover process at this point is as follows:

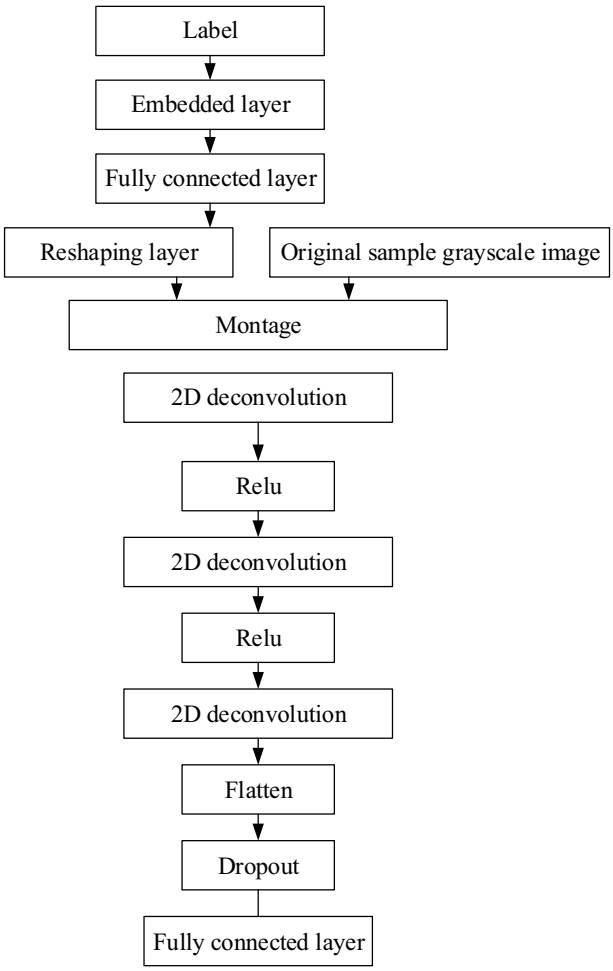

Fig. 2. Discriminator flow chart

$$u_{i,t}^{j} = \begin{cases} v_{i,t}^{j}, rand(0,1) \leqslant CR \neg or \neg j = j_{rand} \\ x_{i,t}^{j}, rand(0,1) > CR \neg or \neg j \neq j_{rand} \end{cases} \tag{5}$$

The selection process is as follows:

$$X_{i,t+1} = \begin{cases} U_{i,t}, f(U_{i,t}) \geqslant f(X_{i,t}) \\ X_{i,t}, f(U_{i,t}) < f(X_{i,t}) \end{cases} \tag{6}$$

where $X_{b,t}$ is the optimal individual in the population, $V_{i,t}$ is the newly generated variant individual,$F \in [0,2]$ is the variance factor, $j_{rand} \in \{1,2,\ldots,s\}$ is the index number of the individual vector dimension,$CR \in (0,1)$ is the crossover probability factor, and $X_{i,t+1}$ is the updated individual. The DE strategy updates the genes of the mixed population through mutation, crossover, and selection. In the AWA-CWGAN algorithm, this strategy is applied during the global search phase to optimize the weights of the generator and discriminator. Combined with CWGAN's generative capabilities, differential evolution enables more effective exploration of the parameter space, thereby enhancing the model's predictive performance.

When combining the differential evolution strategy with CWGAN for price prediction, issues such as slow convergence, suboptimal solution accuracy, and low operational efficiency still persist. To address these challenges, this paper proposes the AWA-CWGAN algorithm, which integrates adaptive weight adjustment with the differential evolution strategy to enhance performance.

To begin, the algorithm uses neighborhood distance to evaluate the sparsity of the current solution. This sparsity is calculated as the average Euclidean distance of an individual to its nearest neighbors in the population. Sparsity is used to measure the distribution of individuals within the population and is calculated as the average Euclidean distance between an individual and its nearest neighbors. The weight adjustment strategy dynamically modifies an individual's weight based on its sparsity to maintain population diversity. When the sparsity is low—indicating the presence of many similar individuals nearby—the weight is reduced to prevent excessive exploitation. Conversely, when the sparsity is high—implying fewer similar individuals nearby—the weight is

increased to encourage further exploration. The sparsity of the -th individual in the population is defined as follows:

$$S_j = \frac{\sum_{i=1}^{m} L^{NN_i^j}}{m} \tag{7}$$

where$L^{NN_i^j}$ is the Euclidean distance from the jth solution to the ith nearest neighbor and m is the number of objective functions.

Based on the sparsity of all individuals, subproblems corresponding to individuals with sparsity lower than a predefined threshold are removed, while new subproblems are introduced in regions with high sparsity. To achieve this, a weight vector is added between the subproblem associated with the individual having the highest sparsity and its nearest neighbor's corresponding subproblem. Let and represent the weight vectors of the two subproblems. The new weight vector corresponding to the subproblem is calculated as follows:

$$W = (\tfrac{u_1+v_1}{2}, \tfrac{u_2+v_2}{2}, \ldots, \tfrac{u_m+v_m}{2}) \tag{8}$$

A solution from the external population that minimizes the aggregation function's value with the newly assigned weights is selected as the solution to the new subproblem. Figure 3 shows that the external population retains the non-dominated solutions that have been eliminated by the main population. Selecting a new solution from the external population primarily aims to ensure population diversity while also considering convergence.

The AWA-CWGAN algorithm builds upon CWGAN by incorporating the AWA mechanism and a differential evolution strategy. The AWA mechanism dynamically adjusts weights based on the evolutionary state of the population and the fitness of individuals, aiming to preserve population diversity and guide the search direction. The differential evolution strategy further refines the genetic structure of the population through mutation, crossover, and selection. By integrating these two strategies, AWA-CWGAN effectively balances exploration and exploitation during the global search process, thereby improving the accuracy of price prediction.

Experimental analysis

In this section, experimental simulations are conducted using the TensorFlow-GPU 2.4.2 deep learning framework on a Windows 11 operating system. The computer configuration includes an AMD Ryzen 7 6800 H processor with Radeon Graphics (3.20 GHz) and 16 GB of memory. Within this experimental setup, the performance of the proposed prediction algorithm, AWA-CWGAN, is analyzed. The autoencoder component consists of two fully connected layers, each containing 128 neurons and utilizing the ReLU activation function. The latent dimension is set to 64 to effectively capture key features of the input data. The LSTM component comprises a single layer with 128 neurons, designed to process temporal sequences. The model is optimized using the Adam optimizer with a learning rate of 0.001 to ensure stable training and fast convergence.

Experimental data

The datasets utilized in this study for e-commerce price prediction include Fashion-MNIST and Product-Item-Data. The data was divided into the training set, the validation set and the test set, with proportions of 70%, 15% and 15% respectively.

The Fashion-MNIST dataset consists of 70,000 grayscale images of size 28 × 28, categorized into 10 classes. Each image is associated with a price label, which is generated based on a combination of market research and product characteristics. The Product-Item-Data dataset contains over 100,000 anonymized orders made on Olist across various Brazilian markets between 2016 and 2018. The price data in this dataset has been preprocessed to remove outliers and noise. The Product-Item-Data dataset, on the other hand, comprises over 100,000 anonymized Brazilian orders made on Olist between 2016 and 2018 across multiple marketplaces. It encompasses various dimensions such as order status, price, payment and shipping performance, and real customer reviews. The data files, originally in JSON format, were parsed and converted into a tabular format suitable for analysis. Before being input into CWGAN, the price and attribute features undergo normalization processing, and the features are scaled to the range of [0, 1] using mine-max normalization. The category features are transformed using exclusive hot coding so that the model can handle them correctly.

When applying SMOTE to time series data, special attention should be paid to maintaining the temporal order. One approach is to first sort the data by time, and then for each minority class sample, find the K nearest neighbors in time for interpolation to generate new samples. This method ensures the temporal continuity of the newly generated samples and avoids disrupting the temporal structure of the original data.

Valuation criteria

In this study, Precision, Accuracy, Recall, and F1-score are employed to evaluate the performance of the price prediction models. These evaluation metrics are derived from the confusion matrix and are calculated using the following formulas:

$$\mathrm{A}cc = \frac{TP + TN}{TP + FP + TN + FN} \tag{9}$$

$$\mathrm{Pr}\,e = \frac{TP}{TP + FP} \tag{10}$$

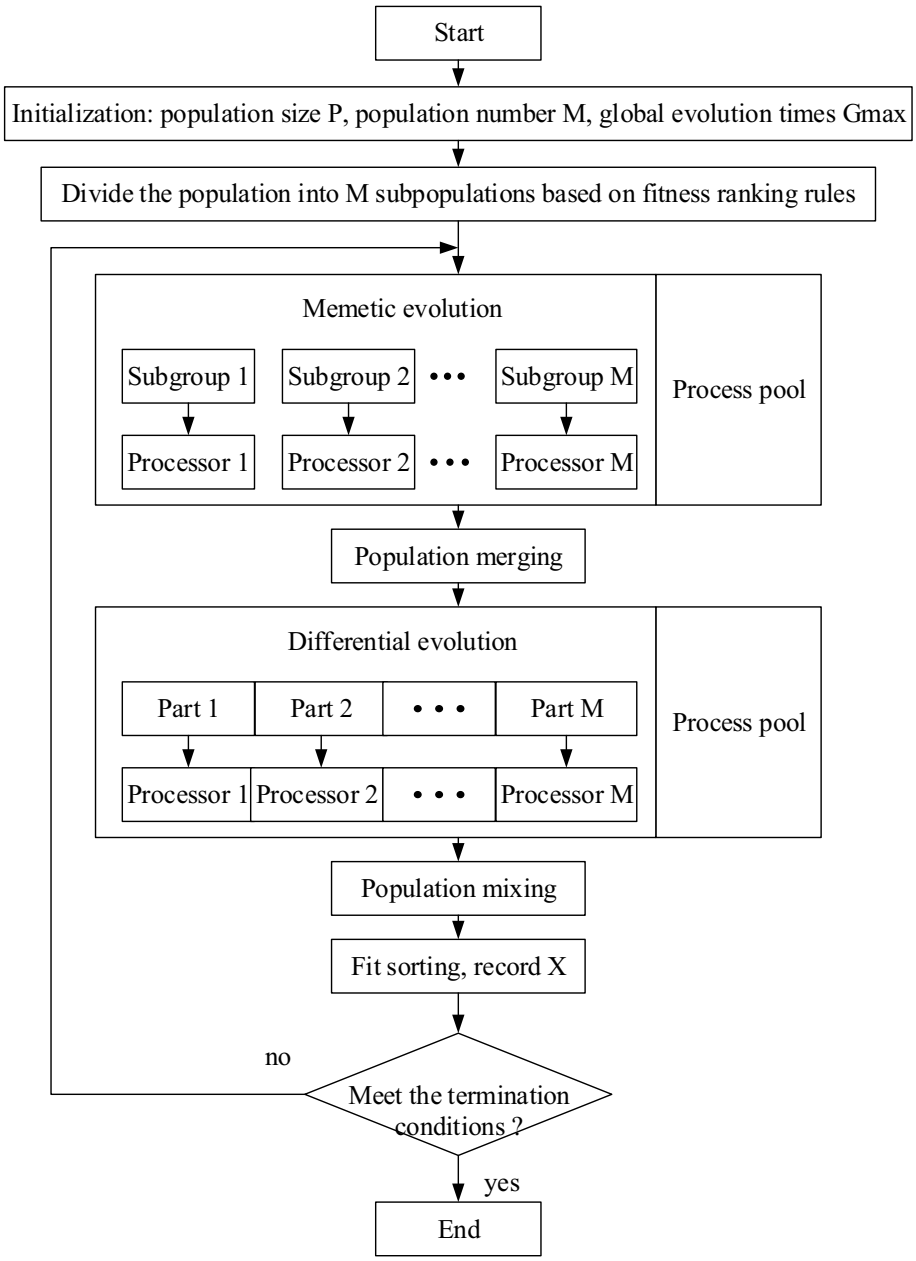

Fig. 3. Evolutionary algorithm execution process

$$\mathrm{Re}\,c = \frac{TP}{TP + FN} \tag{11}$$

$$F1 = \frac{2 \times \mathrm{Re}\,c \times \mathrm{Pr}\,e}{\mathrm{Re}\,c + \mathrm{Pr}\,e} \tag{12}$$

where TP and TN denote the number of samples with upward price forecasts of upward and downward types, respectively, and FP and FN denote the number of samples with upward price forecast flows of downward and upward types, respectively.

Algorithm solving performance analysis

To ensure the comparability of algorithm performance, we compare the proposed method with the existing DE algorithm[22] and the SFLA. The population size is set to 400, with 20 subpopulations and 20 individuals in each subpopulation. The number of global evolutionary generations is fixed at 1,000, the number of local searches is set to 20, and the maximum movement step is defined as the difference between the extreme values of the optimization variables for the corresponding functions. For the Differential Evolution algorithm, the population size is also set to 400, the number of global evolutionary generations is 1,000, and both the variation factor and

crossover probability factor are set to 0.5. In DE, FFF and CRCRCR are critical parameters that control the balance between exploration and exploitation. Specifically, the value of FFF affects the mutation step size—larger values of FFF facilitate global exploration, while smaller values enhance local exploitation. The CRCRCR parameter determines the probability of crossover, where higher values promote information exchange among individuals, thereby helping maintain population diversity. Setting both FFF and CRCRCR to 0.5 is considered a moderate choice, aimed at striking a balance between global search and local refinement, and avoiding premature convergence to local optima. All algorithms terminate after completing 1000 global evolutionary iterations.

During the hyperparameter optimization process of the Walrus algorithm, several key parameters are tuned. The learning rate is searched within the range of [0.0001, 0.01] to balance training stability and convergence speed. The batch size is explored within the range of [32, 256] to evaluate the impact of different data loading strategies on the model's generalization capability. The number of LSTM layers is limited to [1, 2] to determine the optimal model depth for processing temporal sequences. Additionally, the number of LSTM neurons is searched within [64, 256] to adjust the model's ability to capture temporal features. To ensure both efficiency and adequacy of the optimization process, the maximum number of iterations is set to 100, aiming to obtain a reasonably optimal combination of hyperparameters within an acceptable time frame.

Figure 4 illustrates the convergence curves of the three algorithms: DE, SFLA, and AWA-CWGAN. As shown in Fig. 4, the AWA-CWGAN algorithm demonstrates a significantly faster convergence speed during the early stages of evolution compared to DE and SFLA, reflecting its superior capability in locating the optimal solution. In the later stages of evolution, while all three algorithms tend to converge, the proposed algorithm continues updating the solution in the optimal direction until the global optimum is achieved. In contrast, the optimization results of DE and SFLA remain largely unchanged after their initial convergence, indicating limited improvement thereafter.

Notably, AWA-CWGAN maintains sufficient population diversity even in the late stages of global evolution, effectively balancing its exploration and exploitation capabilities and avoiding premature convergence to local optima. Based on the number of global evolutionary generations () and the corresponding fitness values () at stagnation, it is evident that, under identical conditions, the AWA-CWGAN algorithm requires only 16–25% of the global evolutionary generations needed by DE or SFLA to achieve full convergence. Moreover, its convergence accuracy is significantly higher, demonstrating its superior search efficiency.

The overhead performance of the AWA-CWGAN algorithm is analyzed as follows. During testing, the algorithm employed varying numbers of processes for parallel computation, with each configuration executed independently 20 times to ensure result stability and reliability. Notably, when a single process is used, the algorithm operates in serial mode. For parameter settings, the number of subpopulations was set to 60, with 20 individuals per subpopulation, 20 local evolutionary generations, and 1000 global evolutionary generations, ensuring the algorithm runs within reasonable computational resources.

The execution time of the algorithm, depicted in Fig. 5, illustrates its runtime under different numbers of processes. Additionally, Fig. 6 presents the parallel performance metrics of the algorithm. Analysis reveals a marked reduction in the algorithm's runtime as the number of processes increases, accompanied by a gradual rise in the speedup ratio. This demonstrates the effectiveness of parallel computing in enhancing the algorithm's efficiency. Specifically, when the number of processes is below 60, the speedup ratio shows a significant increase, and the parallel efficiency remains relatively high. This indicates efficient utilization of system resources and minimal data communication overhead between processes, enabling effective parallel computation.

However, when the number of processes exceeds 60, the increase in the speedup ratio slows down, and parallel efficiency diminishes. This decline is primarily attributed to higher system resource consumption and increased inter-process communication overhead as the number of processes grows. These additional costs

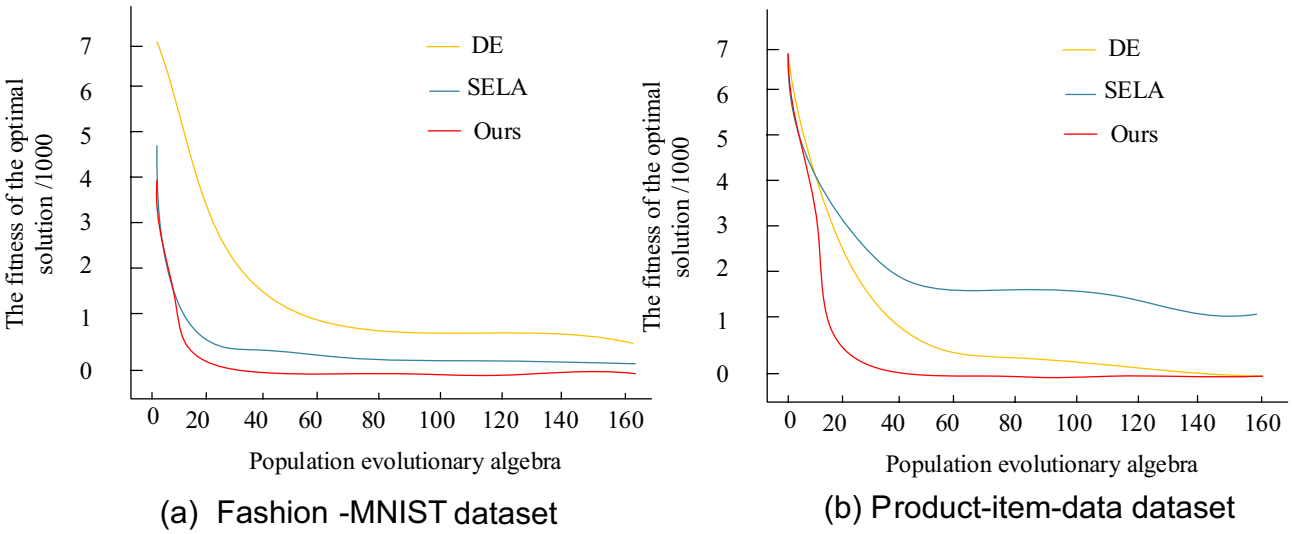

Fig. 4. Comparing the performance of the optimal solution

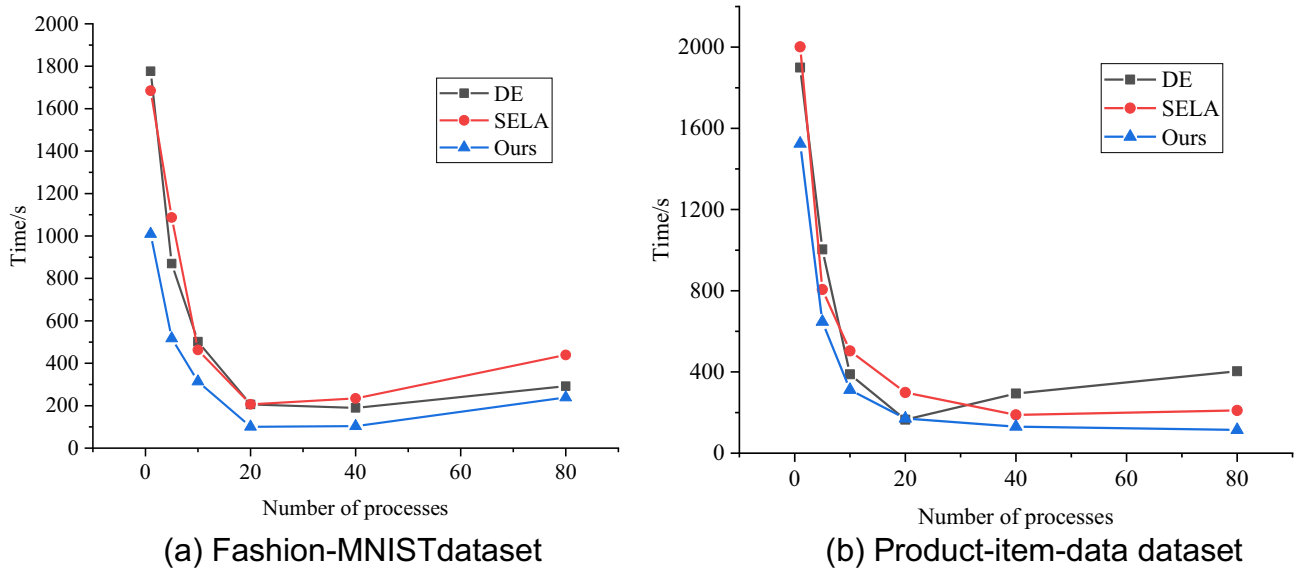

Fig. 5. Computation cost comparison

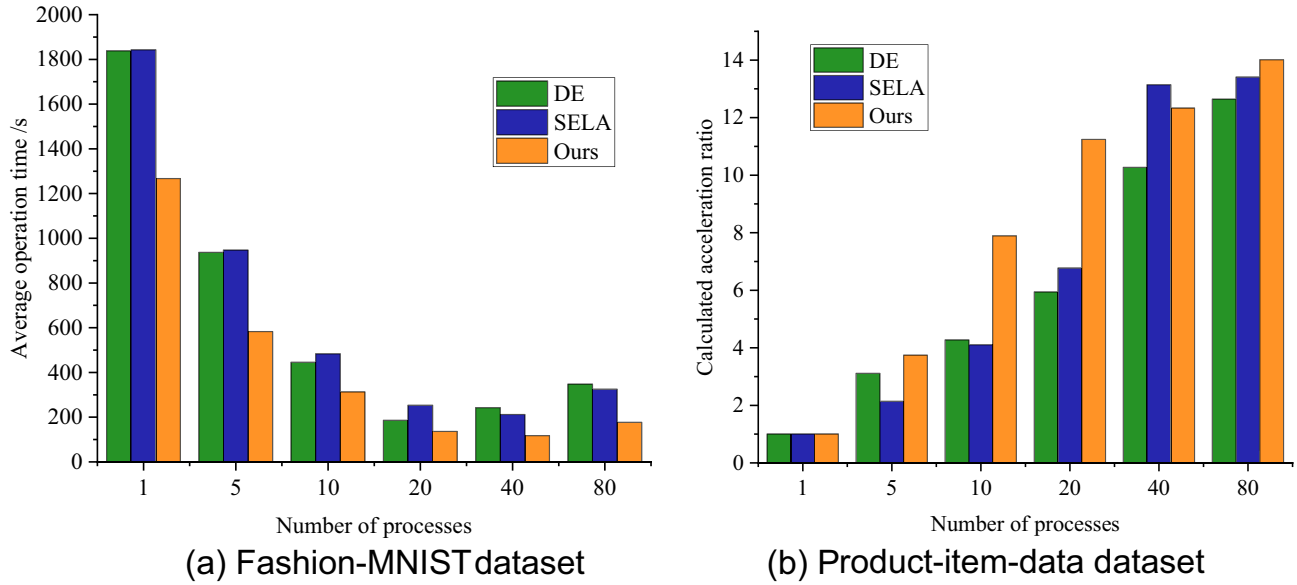

Fig. 6. Comparison of algorithm performance indicators

directly reduce the algorithm's operational efficiency, thereby diminishing the benefits of parallel computing in this scenario.

In summary, the AWA-CWGAN algorithm effectively leverages the advantages of parallel computing to achieve efficient operation within an optimal range of processes. However, when the number of processes becomes excessively large, the increased consumption of system resources and the overhead associated with data communication constrain the algorithm's performance improvement. Therefore, in practical applications, it is crucial to determine the number of processes based on the available computational resources and specific requirements to achieve optimal algorithm performance.

Comparison of predicted performance

The performance results of the three algorithms in price prediction are analyzed as follows, as shown in Fig. 7.

The proposed method achieves accuracy, precision, recall, and F1 scores of 88.8%, 88.81%, 89.255%, and 87.95%, respectively, outperforming the other compared methods. This superior performance is attributed to the limitations of the comparison algorithms, which fail to address the dataset imbalance problem in e-commerce product price prediction and rely solely on resampling the original data. In contrast, the proposed algorithm integrates Wasserstein scattering into CWGAN while removing Lipschitz constraints, thereby mitigating model

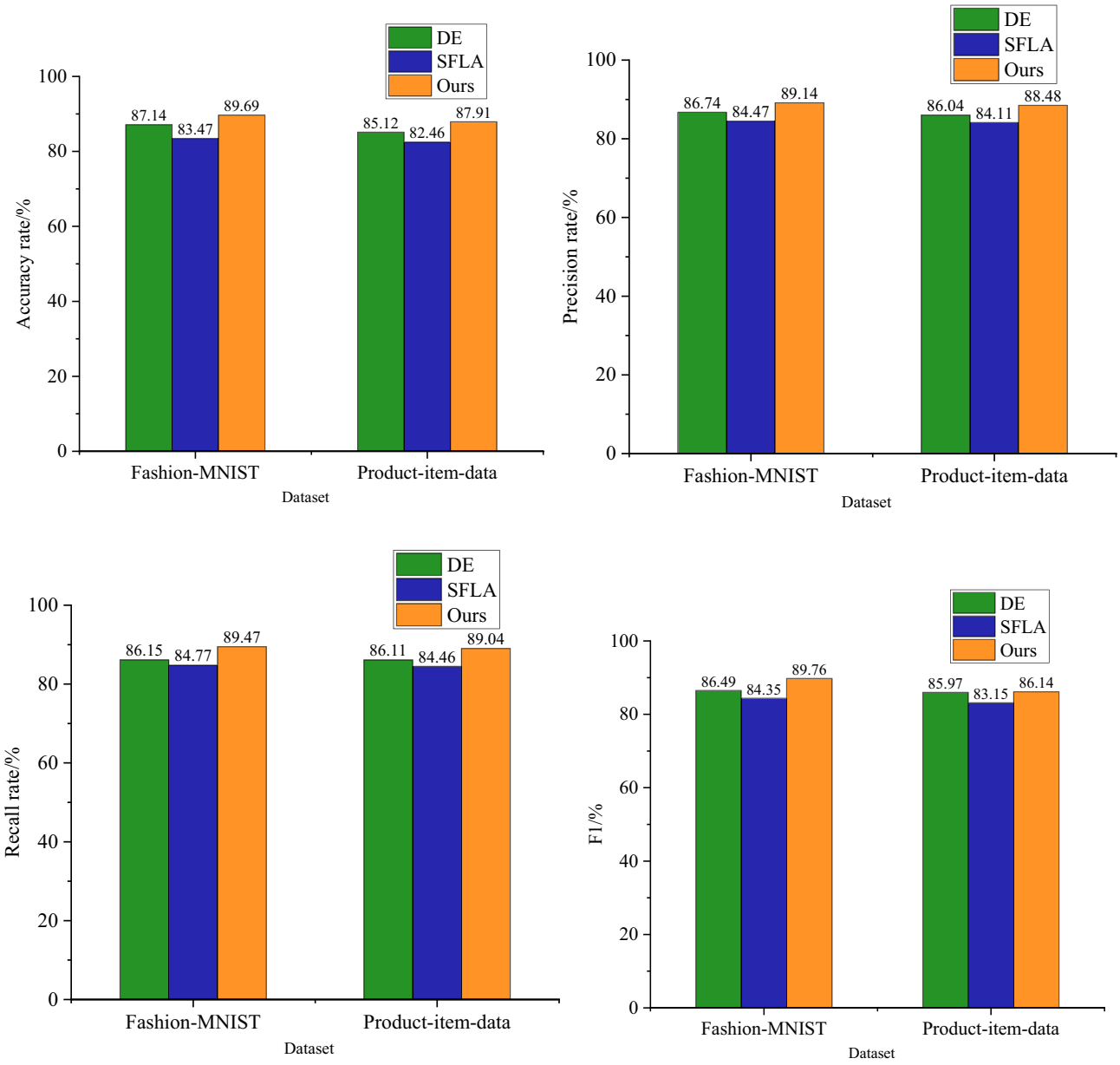

Fig. 7. Ethical decision performance comparison

collapse and training instability issues commonly encountered with WGAN during training. Furthermore, the adaptive weight adjustment mechanism dynamically modifies the weights based on the evolutionary state of the population and the adaptability of individuals. This ensures population diversity is maintained throughout the search process and effectively guides the search direction. By dynamically adjusting the weights, the algorithm avoids premature convergence, such as falling into local optima, thereby enhancing the likelihood of identifying the global optimal solution.

The AWA-CWGAN algorithm successfully combines the benefits of adaptive weight adjustment and the differential evolutionary strategy. This integration allows the algorithm to efficiently balance exploration and exploitation during the global search process, significantly improving its predictive performance.

Conclusion

Traditional GANs often generate simplistic and redundant samples, which fail to adequately capture the complex distributional characteristics of e-commerce product price data. This limitation hampers the model's predictive accuracy and generalization in practical applications. To address these challenges, this paper combines the strengths of CGAN and WGAN to propose the CWGAN model. By improving the quality of newly generated samples, the CWGAN model mitigates issues such as model collapse and training instability during the training process. Building on this foundation, the AWA-CWGAN algorithm is introduced, incorporating an adaptive weight adjustment mechanism and a differential evolution strategy. The algorithm employs a neighborhood

learning strategy to update the optimal individuals within subpopulations and dynamically adjusts the weights based on data sparsity. This adaptive approach ensures the model maintains population diversity and effectively guides the optimization process. As a result, the AWA-CWGAN algorithm demonstrates superior performance compared to traditional methods, achieving accuracy, precision, recall, and F1 scores of 88.8%, 88.81%, 89.255%, and 87.95%, respectively. While existing adaptive weight adjustment strategies have achieved notable success in balancing diverse data distributions and task requirements, there is significant potential for further optimization. Future research will focus on refining these strategies by integrating more contextual information or advanced feature engineering to enhance the precision of weight adjustments. These enhancements aim to better adapt the model to complex scenarios in e-commerce product price prediction. Additionally, the application of adaptive weight adjustment mechanisms will be explored in other domains, such as financial forecasting and recommender systems, to extend their impact and utility.

Data availability

The datasets generated and/or analysed during the current study are available. Fashion-MNIST: https://github.com/zalandoresearch/fashion-mnist, doi: https://doi.org/10.48550/arXiv.1708.07747, Product-Item-Data: https://www.kaggle.com/datasets/cclark/product-item-data, doi: https://doi.org/10.1145/3556089.3556149.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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